Ayush Singh

SOC Intern | Tier 1 Security Analyst | Cybersecurity Undergraduate
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Education

VIT-AP University Amaravati, India
B.Tech in Computer Science Engineering – Cybersecurity Specialization *2023 – 2027*
- Relevant coursework: Network Security, Operating Systems, Data Structures and Algorithms, DBMS, and Software Engineering.
- Actively develop cybersecurity lab skills and maintain security-focused GitHub projects involving Python automation, log analysis, network security, and threat intelligence.

Technical Skills

SOC & Blue Team: SIEM Fundamentals, Wazuh, Splunk, Log Analysis, Incident Response, Phishing Triage, IOC Enrichment, IDS, MITRE ATT&CK, Sysmon
Security Tools: Wireshark, tcpdump, Suricata, Snort, Nmap, Burp Suite, Metasploit, Procmon, Sysinternals, FakeNet-NG
Malware Analysis: REMnux, PEStudio, PE-bear, Ghidra, x64dbg, Regshot, Cuckoo Sandbox Fundamentals
Programming: Python, Bash, JavaScript, Java, C/C++, SQL; scripting, automation, JSON, REST APIs
Systems & DevSecOps: Linux, Kali Linux, Ubuntu, Windows Event Logging, Docker, Git/GitHub, CI/CD, VirtualBox, VMware

Security Projects

SIEM Endpoint Detection & Threat Hunting Lab | *Wazuh, Sysmon, MITRE ATT&CK* 2026
- Explore endpoint monitoring and SIEM workflows using Windows Sysmon telemetry and virtualized security lab environments.
- Practice investigating suspicious authentication and credential-access activity, interpreting security events, and mapping relevant behaviors to MITRE ATT&CK.

ThreatLens – Automated IOC Enrichment | *Python, VirusTotal, AbuseIPDB, REST APIs* 2026
- Developed a modular CLI tool to validate IP addresses and file hashes, query VirusTotal and AbuseIPDB, and normalize threat-intelligence results.
- Implemented risk assessment, SQLite caching, structured JSON/CSV reporting, and optional webhook notifications to support repeatable IOC triage.

Malware Network Traffic Analysis | *PCAP, Wireshark, Suricata* 2026
- Practice packet-level investigation using PCAP files to identify suspicious endpoints, network indicators, unusual connections, and potential command-and-control patterns.
- Organize findings into incident-style notes covering observed indicators, investigation evidence, potential impact, and recommended containment steps.

Security Monitor & Log Analyzer | *Python, NIST, CLI, CI/CD* 2025
- Built a Python-based firewall and server log analyzer with security checks, monitoring workflows, and incident-tracking functionality.
- Implemented CLI-driven analysis and Markdown report generation to organize findings for simulated Tier 1 investigations and documentation.

Email Phishing & Spoofing Analyzer | *Python, Flask, REST API, DevSecOps* 2025
- Developed an `.eml` analysis tool to inspect email headers, flag suspicious sender information, evaluate spoofing indicators, and extract potentially malicious URLs.
- Used structured parsing and automated testing workflows to improve repeatability and support safer analysis of suspicious email samples.

Experience

Vectr Hyderabad, Telangana
Web Developer – Secure Development & API Engineering *Sept. 2023 – Jan. 2026*
- Worked with FastAPI, Docker, and REST APIs, contributing to API implementation, application reliability, and secure development practices across deployment workflows.
- Supported Git-based collaboration, code maintenance, and deployment pipelines, applying structured development workflows to improve consistency and reduce implementation issues.
- Contributed to production software maintenance and troubleshooting; published a community plugin with 2,000+ downloads and a 4.5/5-star rating.

CTFs, Security Labs & Certifications

CTF Platforms: OverTheWire Bandit (completed through Level 33), Natas web-security challenges, TryHackMe, Hack The Box, VulnHub
Linux & Web Security Labs: SSH-based challenges, file permissions, Linux enumeration, command-line text processing, HTTP requests, authentication, source inspection, and web vulnerability investigation using curl and Burp Suite
Network & Malware Labs: Nmap reconnaissance, Wireshark packet inspection, tcpdump traffic capture, Windows/ Linux virtual machines, REMnux analysis environment, FakeNet-NG network simulation, Procmon activity monitoring, and Regshot change comparison
Security Practice: Python security scripting, DNS and network troubleshooting, vulnerability assessment fundamentals, phishing analysis, and incident-response documentation
Certifications: Cisco Introduction to Cybersecurity; NPTEL Cybersecurity

Leadership

Founding Partner & Treasurer, Entrepreneurship Club, VIT-AP: Supported club operations, budgeting, sponsorship, and event coordination.

Co-Founder, VITAURA Technical Fest: Contributed to event planning, team coordination, and outreach across university technical tracks.